

Project Report: Research Agent with Web Search - Agentic AI Engineer

Written by Gagan Pasupuleti

Family:	Agentic AI Engineer
Level:	Beginner
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

LangChain agent that searches the web, summarizes findings, and cites sources.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Beginner]

Chapter focus: Project Overview** *(Level: Beginner)

Research Agent with Web Search** is a beginner-level end-to-end project for Agentic AI Engineer. **Duration:** 2 weeks. **LangChain agent that searches the web, summarizes findings, and cites sources.** **Tools:** LangChain, Python, Search API.

Code Reference:

Objective: Define 3 tools (search, summarize, save)

Objective: Agent loop with max 5 steps

Objective: Output markdown report with citations

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Agentic AI Engineer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Agentic AI Engineer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Tools [Beginner]

Chapter focus: Phase: Tools** *(Level: Beginner)

Implement tool functions. Complete these tasks: Web search API; File writer. Deliverable for this phase: Tool schemas.

Code Reference:

- [] Web search API
- [] File writer

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Tools is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Agentic AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Tool schemas to the teacher.

Chapter 3: Phase: Agent [Beginner]

Chapter focus: Phase: Agent** *(Level: Beginner)

ReAct agent loop. Complete these tasks: System prompt; Step limit. Deliverable for this phase: Agent running.

Code Reference:

- [] System prompt
- [] Step limit

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Agent is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Agentic AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Agent running to the teacher.

Chapter 4: Phase: Output [Beginner]**Chapter focus: Phase: Output** *(Level: Beginner)**

Markdown report. Complete these tasks: Citations; Summary sections. Deliverable for this phase: Sample research report.

Code Reference:

- [] Citations
- [] Summary sections

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Output is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Agentic AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Sample research report to the teacher.

Chapter 5: Phase: Test [Beginner]**Chapter focus: Phase: Test** *(Level: Beginner)**

5 benchmark queries. Complete these tasks: Accuracy check; Source verify. Deliverable for this phase: Test results doc.

Code Reference:

- [] Accuracy check
- [] Source verify

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Test is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Agentic AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Test results doc to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Beginner]

Chapter focus: Final Deliverables and Review *(Level: Beginner)**

Submit all final deliverables: Agent code; Sample reports; Test results. Skills demonstrated: Agents, Tool use, LangChain. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

Deliverable: Agent code
Deliverable: Sample reports
Deliverable: Test results

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide**What it actually is**

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Agent code and passes the rubric.